Applications of quadratic functions

1 Mae throws a stick vertically upwards. After t seconds, its height h feet above the ground is given by the formula $h = 25t - 5t^2$.

- **a** At what time(s) will the stick be 30 feet above the ground?
- **b** How long does the stick take to hit the ground?
- **c** Can the stick ever reach a height of 36 feet?

2 A t-shirt is fired straight up from a t-shirt cannon at ground level. After t seconds, its height above the ground is h meters, where $h = -16t^2 + 48t$.

- **a** For what values of t is the shirt 11 meters above the ground?
- **b** Calculate how long the t-shirt was at least 11 meters above the ground.
- **c** For what values of t is the shirt 35 meters above the ground?
- **d** Calculate how long the t-shirt was at least 35 meters above the ground.

3 The area of a rectangle is 160 yd^2. The length of the rectangle is 6 yd longer than its width. Let w be the width of the rectangle in yards.

- **a** Find the width of the rectangle, w.
- **b** Find the rectangle's length.

4 At time t seconds, the distance, s, traveled by an object moving in a straight line is given by

$$s = ut + \frac{1}{2}at^2$$

where u is its starting speed and a is its acceleration.
When $u = 16$ and $a = 8$, find how long it would take for the object to travel 128 yards.

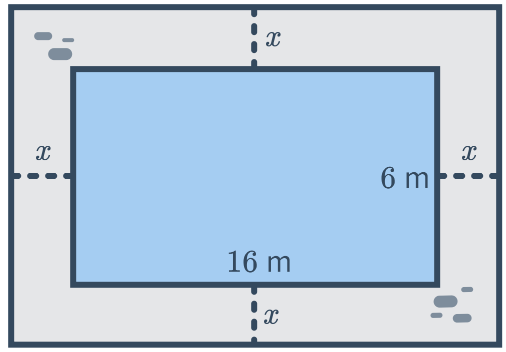
5 The square of a number is 70 less than 17 times the number. Let x be the number.

- **a** What is the largest possible number that satisfies this condition?
- **b** Verify that the difference between the square of the number and 17 times the number is 70, for this value of x.

6 The sum of the areas of two circles is $79.56\pi \text{ in}^2$. One of the circles has a radius of 1.1 times the length of the other. Let r be the length of the shorter radius.

- **a** Find the length of the shorter radius.
- **b** Find the length of the longer radius.

7 A rectangular swimming pool is $16 \mathrm{~m}$ long and $6 \mathrm{~m}$ wide. It is surrounded by a pebble path of uniform width $x \mathrm{~m}$. The area of the path is $104 \mathrm{~m}^2$.



a Find an expression for the area of the path in terms of x. Express your answer in simplest form.

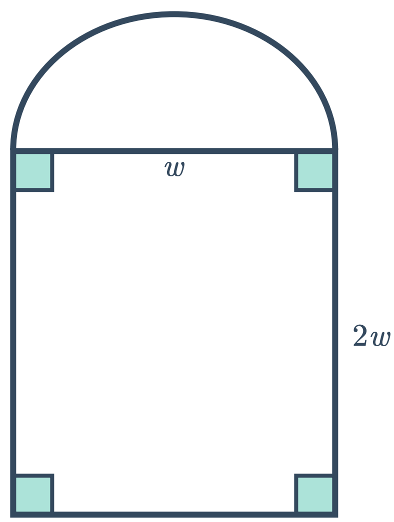
b Now, form an equation and solve for x, the width of the path.

8 Georgia can break a code in 5 hours less time than her younger sister Tricia. If they work together, the code will take them 6 hours to break.

a Find the number of hours, m, that it would take Georgia to break the code by herself.

b Now, find how long it would take Tricia to crack the code by herself.

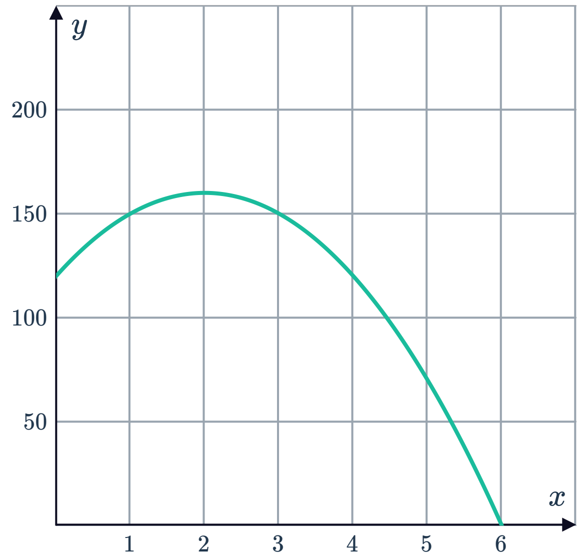
9 The diameter of a semicircle shares its edge with the width of a rectangle whose length is twice its width. Their combined area is $13 \mathrm{~ft}^2$.



a Find the width of the rectangle correct to two decimal places. Let w be the width.

b Now, find the perimeter of the entire shape correct to one decimal place.

10 A frisbee is thrown upward and and away from the top of a mountain that is 120 meters tall. The height, y, of the frisbee at time x in seconds is given by the function $y = -10x^2 + 40x + 120$. This function is graphed below.



a What is the y-value of the y-intercept of this graph?

b What does the value of the x-intercept represent in context?

c Given that the x-value of the vertex is 2, what is the maximum height reached by the frisbee?

11 A ball is thrown into the air from the top of a cliff. The height, $h(t)$, of the ball above the ground t seconds after it is thrown can be modeled by the following function:

$$h(t) = -5t^2 - 15t + 90$$

How many seconds after being thrown will the ball hit the ground?

12 On Mercury the equation $d = 1.5t^2$ can be used to approximate the distance in meters, d, that an object falls in t seconds, if air resistance is ignored.

a Complete the table:

Time	0	2	4	6
Distance				

b Graph the function $d = 1.5t^2$.

c Use the equation or otherwize to determine the number of seconds, t, that it would take an object to fall 5.6 m. Give the value of t to the nearest second.

13 On Earth, the equation $d = 4.9t^2$ is used to f[illegible]e distance (in yards) an object has fallen through the air after t seconds.

a. 8 seconds after a skydiver jumps out of the plane, what distance has she fallen?

b. Solve for t, the time it will take her to fall 78.4 yards.

14 The formula for the surface area of a sphere is $S = 4\pi r^2$, where r is the radius in inches.

r	1	2	3	4	5	6	7	8
S								

a. Complete the following table of values for $S = 4\pi r^2$, round your answers to two decimal places.

b. Graph the correct curve for $S = 4\pi r^2$

c. From your graph, determine the closest value of the following:

   i. The surface of area of a sphere with radius 5.5 in.

   ii. The radius of a sphere with a surface area of 201 in^2.

15 A compass is accidentally thrown upward and out of an air balloon at a height of 100 feet. The height, y, of the compass at time x in seconds is given by the equation:

$$y = -20x^2 + 80x + 100$$

a. Fill in the following table of values for $y = -20x^2 + 80x + 100$:

x	0	1	2	3	4	5	6
y							

b. Using the table, determine when the compass strikes the ground.

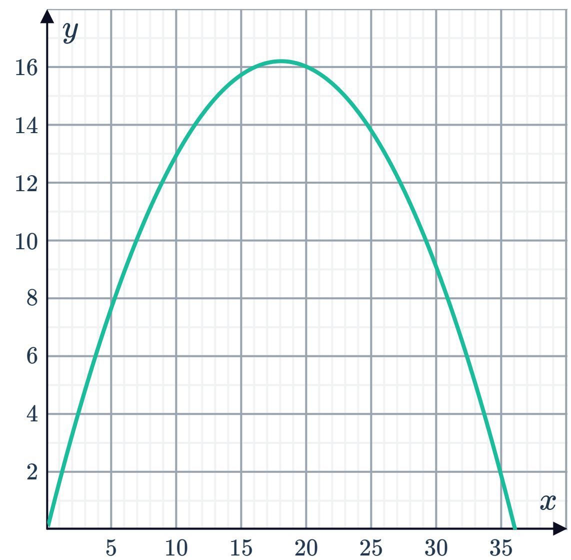
c. According to the table of values, what is the maximum height of the compass?

d. Graph the relationship $y = -20x^2 + 80x + 100$.

16 Large sprinklers are used to water crops. Wh[redacted]ater is sprayed, it takes the path of a parabola. The stream of water reaches a greatest height of 16.2 yards above the ground, 18 yards from the sprinkler.
Let the position of the sprinkler be $(0, 0)$. Let x be the horizontal distance from the sprinkler and y be the height of the water above the ground.

a The graph of the water from the sprinkler is shown. Write down the rule linking x and y.

b If the sprinkler were to rotate 360°, what area of crops would it irrigate? Round your answer to two decimal places.



17 In a game of tennis, the ball is hit high up into the air. Initially the ball is struck 3.5 meters above the ground and hits the ground 7 seconds later. It reaches its greatest height 3 seconds after being hit.

a Determine the coordinates of the points labelled A, B and C in the diagram.

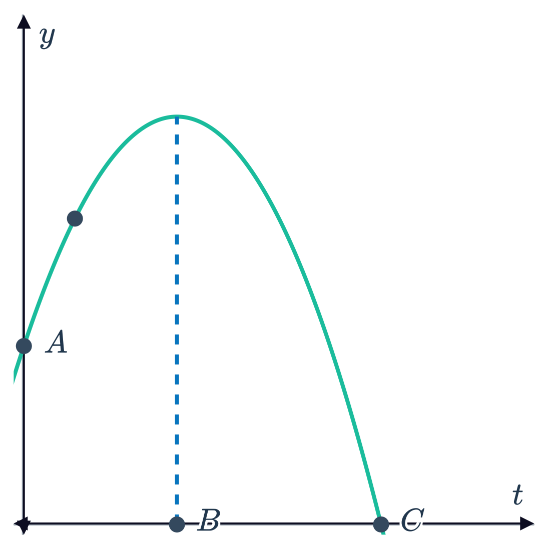
b Using the form of the parabola $y = a\left(t - h\right)^2 + k$, where t is the number of seconds after the ball is hit and y is the height of the ball above the ground, determine the value of h.

c By substituting $t = 0$ and $y = 3.5$ into the equation, form an equation in terms of a and k.

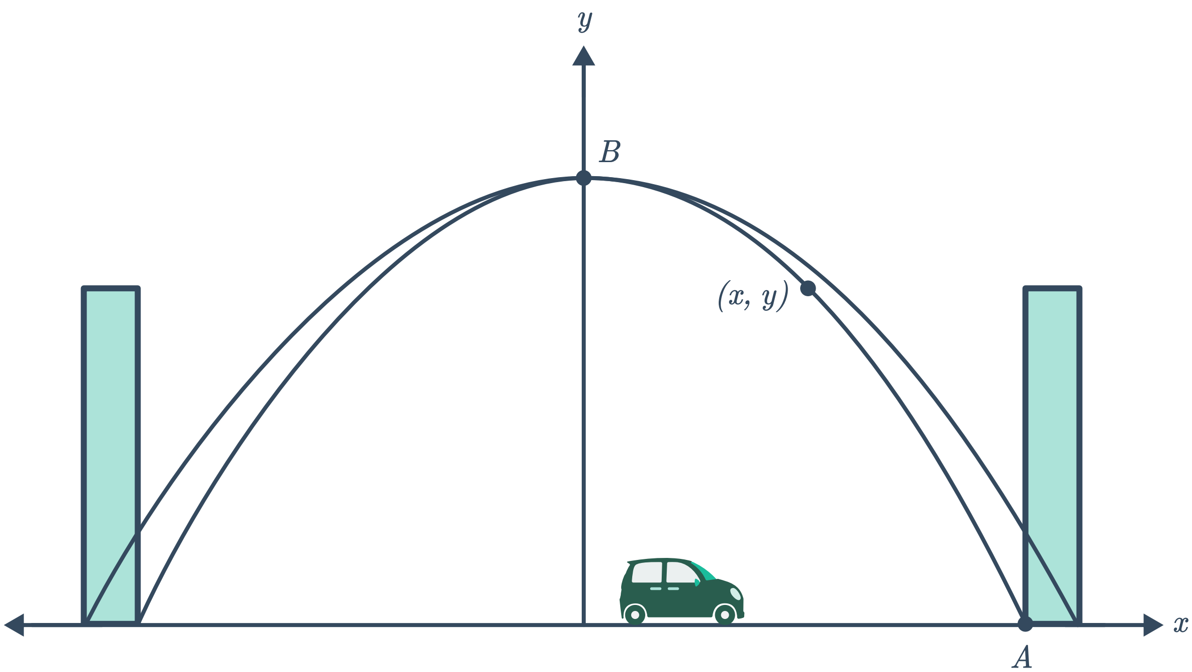
d By substituting $t = 7$ and $y = 0$ into the equation, form another equation in terms of a and k.

e Solve for a.

f Find k, the greatest height reached by the ball.



18 The arch of a parabolic bridge starts 18 yard[redacted]he left of its center and ends 18 yards to the right of its center, and the peak of its arch is 108 yards above the road:



a) Determine the coordinates of the points labelled A and B on the diagram.

b) Find a formula for y, the height of the arch above the road, at distance x yards from the center of the bridge.

c) Pedestrians can climb along the arch up to a vertical point that is half the height of the arch. At their greatest point, how far horizontally are pedestrians from the center of the arch?

19 To minimize the chance of large avalanches, small explosives are fired onto mountain sides. One such explosive is fired at a height of 24 meters and follows a parabolic flight path. It reaches a maximum height of 122 meters, 180 meters horizontally from where it is fired. Let x and y represent the horizontal and vertical distances respectively, of the explosive from where it is fired.

a) Assuming the position where it is fired is $(0, 0)$, determine the vertex of the parabolic path.

b) Using the form of the parabola $y = -a\left(x-h\right)^2 + k$, determine the exact value of a. Then use this value to determine the equation between x and y.

c) The explosive hits a slope that is 361 meters horizontally from the point where it was fired. Does the explosive land above, at the same height or below the point where it was fired?

20 The distance a freely falling object falls is modeled by the formula $d = 16t^2$, where d is the distance in feet that the object falls and t is the time elapsed in seconds.

a Complete the table.

t	0	1	2	3		
d	0	16			400	576

b When $t = 0$, we get $d = 0$. What does this mean?

c If we substitute $d = 400$ into $d = 16t^2$, both $t = 5$ and $t = -5$ would satisfy the equation. Explain why $t = -5$ make sense here?

21 On the moon, the equation $d = 0.8t^2$ is used to approximate the distance an object has fallen after t seconds. (Assuming no air resistance or buoyancy). On Earth, the equation is $d = 4.9t^2$.

a A rock is thrown from a height of 60 meters on Earth. Solve for t, the time it will take to hit the ground to the nearest second.

b If a rock is thrown from a height of 60 meters on the moon, solve for the time, t, it will take to reach the ground to the nearest second.

22 The Gateway Arch in St. Louis, Missouri, is 630 ft tall.
We can model the behavior of objects falling from the arch using Galileo's formula for falling objects: $d = 16t^2$, where d is distance fallen in feet and t is time in seconds since the object was dropped.
How long would it take an object dropped from the arch to fall to the ground? Round your answer to the nearest second.

23 The kinetic energy E of a moving object is given by $E = \frac{1}{2}mv^2$, where m is its mass in kilograms and v is its speed in meters/second.

a If a vehicle travelling at 18 m/s has kinetic energy $E = 307\,800$, what is the mass of the car?

b If a vehicle is travelling at 18 m/s, what is its speed in kilometers/hour to one decimal place?

c If a vehicle weighing 1600 kilograms has kinetic energy $E = 204\,800$, at what speed is it moving?

24 The sum of the series $1+2+3+...+n$ is  by $S=\frac{n(n+1)}{2}$.

a Find the sum of the first 32 positive integers.

b Find the number, n, of positive integers required for a sum of 15.

Maximum and minimum problems

25 The height h, in meters, reached by a ball thrown in the air after t seconds is given by the equation $h=12t-t^2$.

a Fill in the following table of values for $h=12t-t^2$

t	1	2	3	4	5	6	7	8	9	10
h										

b Graph the relationship $h=12t-t^2$.

c Determine the height of the ball after 7.5 seconds have elapsed.

d What is the maximum height reached by the ball?

26 The sum of two integers is 80.

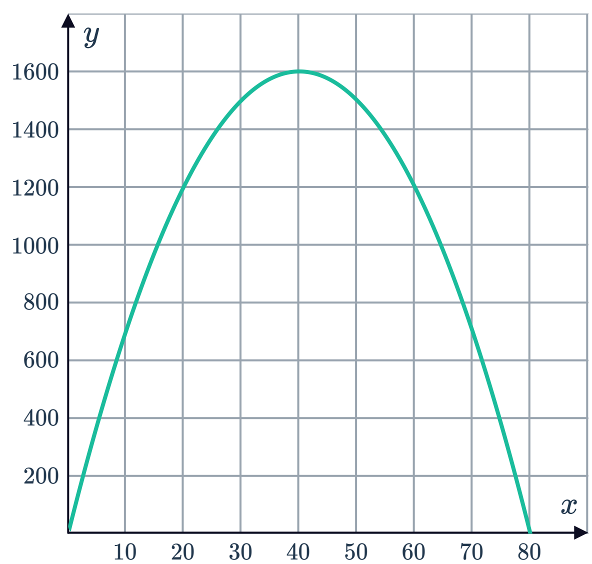
a If one of the integers is 22, what is the other integer?

b Let one of the integers be x. Write an expression for the other integer.

c Let y be the product of the integers. Write an expression for y in terms of x.

d The graph of $y=80x-x^2$ is given. When $x=40$, what is the product of the two integers?

e When $x=40$, does this result in the largest possible product?



27 A rectangle is to be constructed with 80 meters of wire. The rectangle will have an area of

$$A = 40x - x^2$$

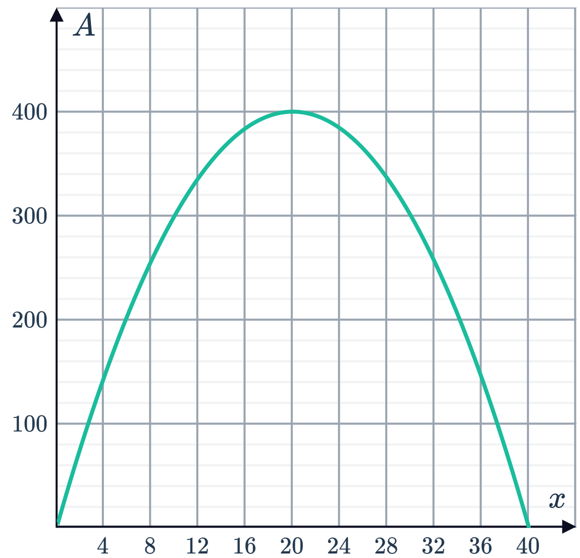
where x is the length of one side of the rectangle. The graph of this function is given:

a Using the equation, state the area of the rectangle if one side is 12 meters long.

b Using the graph, determine the nearest value for the longer side of a rectangle that has an area of 256 square meters.

c Using the graph, what is the greatest possible area of a rectangle that has a perimeter of 80 m?

d Using the graph, state the dimensions of the rectangle with the maximum area.



28 The sum of two whole numbers is 24. Let one of the numbers be x.

a Let y represent the product of the numbers. Form an expression for y in terms of x.

b Find the value of x that will result in the greatest product of the two numbers.

c Find the greatest possible product of the two numbers.

29 A rectangle is constructed such that the sum of its length and width is 20. Let the length of the rectangle be x.

a Form an expression for A, the area of the rectangle, in terms of x.

b What length of rectangle will result in the maximum possible area?

c What is the width of the rectangle with maximum area?

30 When an object is thrown into the air, its height above the ground is given by the equation

$$h = 193 + 24s - s^2$$

where s is its horizontal distance from where it was thrown.

a Find the value of s, at the point when it reaches its greatest height above the ground.

b Find the maximum height reached by the object.

31 The height at time t of a ball thrown upwar[illegible]iven by the equation $h = 59 + 42t - 7t^2$.

a How long does it take the ball to reach its maximum height?

b Find the height of the ball at its greatest point.

32 A ball is thrown upwards from a height of 4 m with an initial velocity of 44 m/s. The height at time t of the ball is given by the equation $h = -16t^2 + 44t + 4$.

a How long does it take the ball to reach the ground? Round your answer to two decimal places.

b How long does it take the ball to reach its maximum height? Round your answer to two decimal places.

c Find the height of the ball at its greatest point, correct to two decimal places.

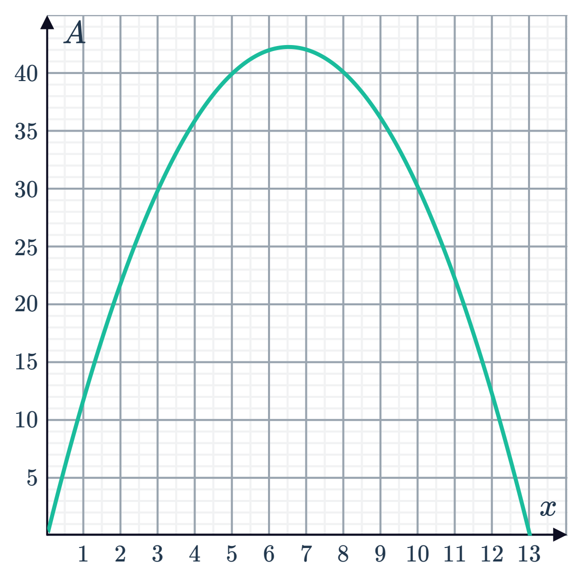
33 A rectangular enclosure is to be built for an animal. Zookeepers have 26 yards of fencing, but they want to maximize the area of the enclosure. Let x be the width of the enclosure. The area function $A = x\left(13 - x\right)$ has been graphed below:

a Form an expression for the length of the enclosure in terms of x.

b Form an expression for A, the area of the enclosure.

c What width will allow for the greatest possible area?

d What is the greatest possible area of such an enclosure?



34 A rectangular paved labyrinth is to be built in the local Botanic Gardens. The stonemasons have 336 m of rare marble for the perimeter, but they want to maximize the area of the labyrinth. Let x be the width of the labyrinth.

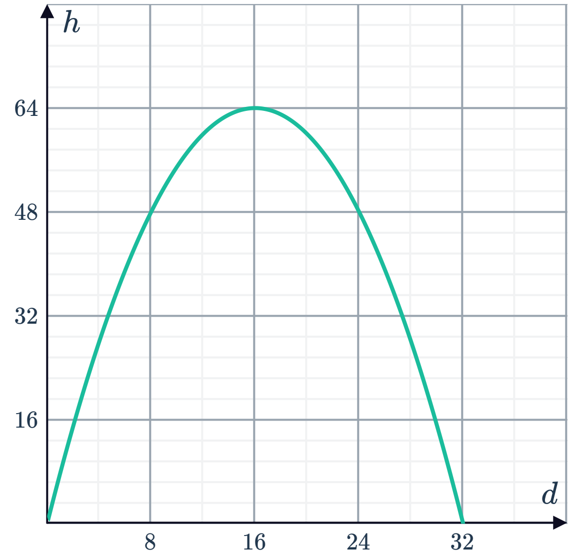
a Find an expression for the length of the labyrinth in terms of x.

b Form an expression for A, the area of the labyrinth.

c What width will allow for the greatest possible area?

d What is the greatest possible area of such a labyrinth?

35 When a fire hose is turned on, the height h [redacted]s of the stream of water is described by $h = -0.25d\,(d - 32)$, where d meters is the horizontal distance from the hose.



a Let the hose be positioned at $(0, 0)$. Use the graph to determine how far from the hose the water reaches its maximum height above ground.

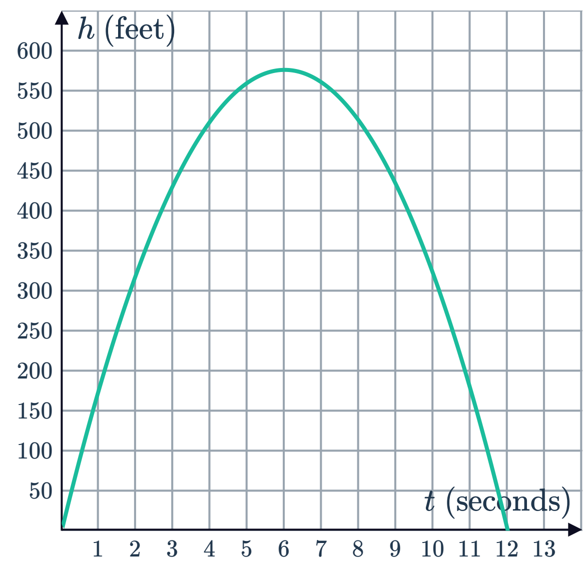
b Using the graph, determine the maximum height reached by the hose.

c There are two points on the path where the water is at a height of 48 meters above the ground. What is the horizontal distance between these two points?

d Determine the height of the water above the ground when it is a horizontal distance of 6 meters from where the fire is.

e For what values of d is the model valid?

36 An object launched from the ground has a height (in feet) after t seconds that is modeled by the graph.

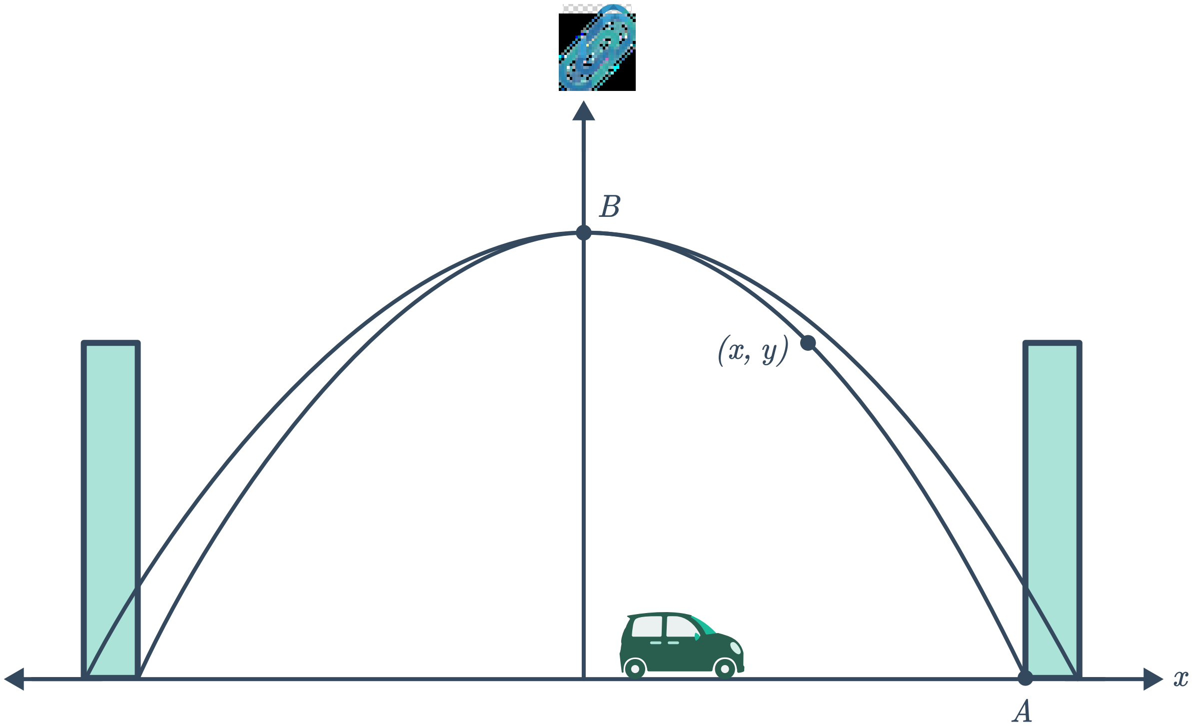


a What is the maximum height of the object?

b After how many seconds is the object at its maximum height?

c How many seconds after launch does the object return to the ground?

37 The arch of a bridge is built so that its height above the horizontal road can be modeled by a parabola whose equation is $y = -\frac{1}{8}\left(x^2 - 400\right)$, where y represents the height of the arch above the road at distance x meters from the center of the bridge.



a Determine the peak height of the arch.

b Pedestrians can climb along the arch up to a point that is horizontally halfway from the center of the bridge. State the possible x values of this point on the bridge.
Write the two values on the same line separated by a comma.

c At this point, how high up are the pedestrians? Round your answer correct to two decimal places.

d For what values of x does this model not work?